



NON-RIGID SUPPORT FOR VIBRATORY EQUIPMENT

PURPOSE

This document establishes guidelines and recommended procedures for the analysis and design of pile/soil supported structural steel or concrete tabletop structures supporting vibrating equipment. Vibrating equipment covered in this guideline consists of harmonically loaded rotating or reciprocating compressors and pumps. The analysis is of the final structure; therefore this guideline can be applied to steel structures that consist of modules transported to a jobsite, or stick-built structures. This guideline is written assuming usage of the Fluor Reference System program SAP2000.

SCOPE

This practice includes information about the following major topics.

- Definitions
- Information Necessary for Design
- Performance Criteria
- Analytical Model
- Dynamic Analysis
- Evaluation of Results
- Additional Topics

APPLICATION

This practice applies to rotating or reciprocating equipment supported on steel structures or modules and to rotating equipment on concrete tabletop foundations. Note that API 686, Chapter 4, Section 4.2 states, "Machinery that requires an elevated installation may be supported on structural steel of adequate stiffness and strength."

The minimum size of equipment for which a dynamic analysis needs to be performed is a judgmental issue. There are rules of thumb applicable for dynamic equipment mounted on rigid concrete block foundations (i.e. reciprocating machines greater than 200 hp, and rotating machines greater than 500 hp; or machines weighing more than 2,300 kg). These rules might not be conservative enough for flexible steel support.

The maximum size of equipment is not dependent on vibration issues. Criteria to determine which vibratory equipment is modularized include total module weight, module size limits, layout clearances, transmission of vibration to other machines on the same interconnected structure, and installed cost.

The design of vibratory equipment on rigid block foundations is covered by Guidelines 000 215 1233, 000 215 1234 and 000 215 1235.



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DEFINITIONS

Steel Structure Support (Module or Stick-Built)

Modules consist of steel structures plus equipment, piping, electrical, and instrumentation that is preassembled at locations remote from the jobsite. Modules are transported as a unit to the jobsite by road, rail, and/or water. Modules are used to reduce project costs or improve schedule by allowing activities to be performed in parallel by transferring jobsite labor to a module assembly yard. Obviously modules are used where jobsite labor expenses are high, usually at remote locations with short construction seasons. In terms of supporting vibratory equipment, steel structures have relatively little mass with stiffness determined by the size and configuration of framing.

Under special circumstances, steel structures are stick built to support vibratory equipment.

Layout: Configurations for steel support can vary widely, however they appear to fall into two categories: (1) Frames that are crane-lifted into place, jacked and rolled from the transporter, or stick-built can have foundation support distributed at regular spacing across the frame, and (2) frames that are moved into place using transporters must leave space underneath the frame for the transporters. Supports for these frames are provided along the sides of the frame, unless the analysis dictates otherwise, in which case special details can be employed to add interior supports after transporter removal.

Machine Mounting: The machine-frame connection consists of tensioned high-strength bolts with either leveling plates, specialty leveling "chocks", or epoxy chocks. Continuous grout pours are not recommended due to potential steel movements during operation. If required, grout pours are typically done after the module is set at the jobsite in order to avoid damage to the grout that could be caused by flexing effects during transportation.

Initial Sizing: A static analysis establishing members that meet strength, deflection, and stability requirements is typically performed first. Thus the starting point for the dynamic analysis is to begin with sizing based on the static analysis. Other initial considerations are:

- 1) Include layout considerations such as equipment size and clearance dimensions for anchor bolts, piping, electrical, maintenance, and operation.
- 2) The supporting frame should meet static rigidity requirements.

Special Issues: Unique to this type of support is an inherent lack of mass, frame strength and rigidity concerns, and flexing frame during transportation.

Concrete Support (Tabletop)

Concrete tabletop structures are reinforced concrete frames used where a compressor needs to be elevated, or has a downward directed discharge. In terms of vibrating machines, concrete tabletops generally have significant mass due to relatively thick



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concrete elements, and considerable stiffness.

Layout: Configuration of a concrete tabletop structure consists of a top slab which supports the machine, several columns, and a base mat or pile cap. The top slab thickness provides mass and rigidity for the machine frame. The base mat provides rigidity for a soil/pile support calculation basis. Vibration isolators, if provided, are normally placed between the base of the top slab and the top of the columns.

Machine Mounting: The machine-top slab connection is the same as provided for concrete rigid block foundations, and consists of tensioned high strength anchor bolts and grout.

Initial Sizing: Traditional rules of thumb can be used for an initial size for the dynamic analysis.

- 1) Include layout considerations such as equipment size and clearance dimensions for anchor bolts, piping, electrical, maintenance, and operation.
- 2) The mat/pile cap should be rigid with respect to the soil. For criteria for establishing thickness, refer to Guidelines 000.215.1233, 000.215.1234, or 000.215.1235.
- 3) The top slab depth should be one fifth of the span between columns.
- 4) The static bearing pressure should be half of the allowable load for soil or pile support
- 5) The total mass of the structure should be 3 times the machine mass for rotary machines, and 5 times for reciprocating machines.
- 6) The mass of the top half of the structure should be greater than machine mass.
- 7) The center of soil/pile resistance should be within 300 mm of all imposed static loading.
- 8) For rotary machines, the mat/pile cap width should be 0.75 times the height from shaft to bottom of machine. For reciprocating machines, the ratio should be 1.5.
- 9) The columns should be stressed nearly equally under static loading.

Special Issues: Typically there is an opening in the top slab for the machine discharge nozzle. Modeling the base slab using an automesh should be considered for ease of making pile & pile cap layout changes.

Design Sequence

The design sequence is basically to determine a trial sizing, perform an analysis, and make sizing adjustments based on analysis results.



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<u>Step:</u>	<u>Description:</u>	<u>Guideline Reference:</u>
1	Gather Data	Information Necessary for Design
2	Determine the Dynamic Force	Attachments 1 and 2
3	Determine Soil/Pile Impedance	Attachment 3
4	Select Proportions	Definitions / Initial Sizing
5	Prepare Dynamic Analytical Model	Analytical Model
6	Perform a Frequency Analysis	Analysis
7	Perform a Forced Response Analysis	Analysis
8	Determine supporting frame rigidity	Support Rigidity Criteria
9	Determine supporting frame strength	Support Strength Criteria
10	Compare Results with Acceptance Criteria	Evaluation of Results
11	Adjust Proportions if Required & Return to Step 5	
12	Verify Adequacy of Anchorage	Anchorage & Leveling

INFORMATION NECESSARY FOR DESIGN

This section identifies all data required for analysis and design. The paragraphs are grouped by the source of the data.

Machine data should be requested from the machine supplier and should include both machine and driver data. Refer to Structural Engineering Specification 000.215.00920: Structural Data For Mechanical Equipment.

Soil information should be requested from the geotechnical consultant. Refer to Civil Engineering Specification 000.210.CVS02010: Geotechnical Engineering Investigation.

Client and project data will be in the form of project specifications, client specifications, or meeting notes.

Basic Machine Data

Operating Speed: This is the frequency or range of frequencies of the dynamic machine forces. For rotating machines, individual rotor speeds should be provided if different from overall machine speed. For reciprocating machines, primary and secondary speeds will be provided.

Outline Drawing: A layout of the machine provides a basic arrangement, dimensions to the cylinders, shaft, and other components.

Anchor Bolts and Layout: Bolt materials and sizes should be provided in order to verify bolt forces and pier dimensions. Whenever possible, anchor bolts for large compressors should be 12 inches minimum from bolt centerline to face of concrete.

Pier Layout: This layout should provide all dimensions and elevations for machine support. If not provided, pier dimensions may be determined using a minimum of 4 inches from the machine base to the face of the pier.



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Jacking Post Locations and Details: This provides the installation methods to properly align machine components.

Machine Forces

Reciprocating Machines: Primary and secondary machine forces are provided for reciprocating machines. Refer to Attachment 2.

Rotating Machines: This data can consist of dynamic force magnitudes, or rotor weights and eccentricities. Refer to Attachment 1. Sometimes a shaft machine tolerance is indicated. If a shaft machine tolerance or offset eccentricities are not provided, the supplier should be informed of the criterion that is being used for dynamic response checks of the support. Forces values are sometime provided at the connecting bolt locations, which can then be resolved at the center of shaft location. Note that some machines, such as centrifuges, have unbalanced forces that are not based solely on the rotor weight and eccentricity. Such machines will have unique acceptable limits on vibration of the support.

Location of Dynamic Forces: This should indicate the location at which reciprocating forces or rotor eccentricities are applied. Normally, this is at shaft bearing locations.

Static Operating Forces: These include rotor torque, thermal expansion, and gas pressure.

Machine Upset Forces: These can include short circuit and blade loss.

Machine Component Data

Component Weights: The motor and compressor/pump are often broken out as separate parts.

Component Center of Gravity: A center of gravity for each machine component should be indicated, usually on the machine outline drawing. Sometimes, one center of gravity is provided for the entire machine.

Component Mass Moments of Inertia: If provided, mass moments about each axis should be used in the vibration analysis. If not provided, mass moments can be estimated based on the size and dimensions of the machine., or deemed to be minor enough to be neglected.

Additional Machine Criteria

Grout Requirements: The machine supplier is the most important source for grout requirements. Supplier criteria must be carefully examined versus client or construction preferences.

Bolt Post Tensioning Criteria: Inappropriate bolt post tensioning can easily cause excessive movement between the machine and base.

Allowable Amplitudes: If provided, use supplier criteria instead of the criteria provided



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in this practice. If not provided, the supplier should be informed of the criterion that is being used for dynamic checks of the support.

Measurement Point Locations: Amplitude measurement locations should be requested if allowable amplitudes are specified by the supplier. If not supplied, amplitudes should be checked, as a minimum, at rotor shaft bearing locations.

Machine Base Frame Rigidity Criteria: Especially for steel support, permissible frame deformation criteria, for pre and post alignment conditions should be requested.

Geotechnical Data

Soil Properties: Unit Weight, Shear Modulus, and Poisson's Ratio values should be provided for each layer. Separate values may be needed for backfill if embedment is to be included. Comparison values are as follows (Typical Soil Data):

Allowable Soil/Pile Capacity: This is a standard component of every geotechnical investigation.

Soil Material Damping: Also called internal or hysteric damping, this is the energy loss within the soil due to interparticle friction.

Construction Recommendations: Also a standard component; recommendations concerning excavation, backfill compaction, and vibration isolation from other foundations may be provided.

Embedment Recommendations: These should be provided to confirm characteristics of backfill materials.

Soil Layer Evaluation: The optional evaluation of soil layers to provide soil stiffness and damping could be computed by the soil consultant. This would be recommended especially if unusual soil conditions are encountered. Geotechnical consultants can also supply a Dyna5/DynaN file containing recommended geotechnical property input. This file can then be edited as appropriate for the analysis of equipment to be used on the project.

Foundation Vibration Analysis: Some soil consultants have the expertise to provide a partial or complete amplitude calculation. This service could become effective if highly unusual soil conditions need to be evaluated or if computer software is not readily available.

Material Properties

Approximate material properties are listed for reference purposes.

Table 1: Static Properties:

Material	Weight Density (kN/m ³)	Poisson's Ratio (-)	Elastic Modulus (GPa)	Shear Modulus (GPa)
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structural steel	77	0.30	200	76.9
concrete (28 MPa)	23.5	0.20	25	10.4

Table 2: Dynamic Properties

Material	Mass Density (kg/m ³)	Axial Wave Velocity (m/s)	Shear Wave Velocity (m/s)	Damping Ratio (%)
structural steel	7850	5050	3,130	2
concrete (28 MPa)	2400	3220	2,080	5

For the dynamic elastic modulus of concrete, the zero strain tangent modulus is sometimes used. This value can be computed from:

$$E_{\text{DYN}} = 1.4 E \quad (1)$$

where,

E = elastic modulus, MPa

E_{DYN} = dynamic modulus, MPa

PERFORMANCE CRITERIA

Machines mounted on non-rigid supports are expected to perform as well as if provided with a rigid block foundation. Thus the performance criteria is the same, however several issues are more pronounced if dealing with a non-rigid support.

Performance criteria from the supplier should override criteria from other sources.

Frequency Criteria

Natural frequencies of the supporting structure indicate where increased amplitudes may occur due to resonance. The resonant range encompasses the natural frequency plus or minus a pre-designated margin. A frequency analysis is performed in order to confirm that the resonant range has been avoided.

Resonant range criterion is provided in the project specification and is typically described as a multiplier to the natural frequency. If supplier, client, or project specification criteria are not provided, a resonant range of concern for significant amplification of response shall be 0.8 to 1.2 time the machine speed.

Non-rigid supports will have as many natural frequencies as there are degrees of freedom. It is not necessary to calculate and compare every natural frequency. Common criterion is to check each natural frequency associated with a mass participation of 5% or more. The 5% MDOF criteria is a rule of thumb that shows only a limited correlation to peak amplitudes. This is because mass participation is computed from to all deflections in the structure, whereas machine design is interested in amplitudes at only a couple of specific locations.



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Amplitude Criteria

Vibration amplitudes are limited for two reasons. First, amplitudes are an indicator of wear and tear on the machine. Operation at higher amplitudes correlates to shorter intervals between required maintenance shutdowns as well as potential damage to the machine. Second, amplitudes near the machine are limited in order to minimize effects on people near a machine.

Amplitude criteria can be provided in terms of displacement, velocity, or acceleration.

$$V_{PEAK} = (\Delta_{PEAK})(\Omega) \quad (2)$$

$$A_{PEAK} = (\Delta_{PEAK})(\Omega^2) \quad (3)$$

where,

A_{PEAK} = peak acceleration, mm/s²

V_{PEAK} = peak velocity, mm/s

Δ_{PEAK} = peak displacement, mm

Ω = machine frequency, rad/s

Amplitude can also be provided in terms of a root-mean-square (RMS) velocity:

$$V_{PEAK} = V_{RMS} \sqrt{2} \quad (4)$$

where,

V_{RMS} = root mean square velocity, mm/s

Normally each axial direction is evaluated separately; however supplier or project criteria may specify limits in terms of a vector sum of the three axial components.

Machine Limits: Lower vibration amplitudes will generally extend the machine maintenance intervals and validate the supplier's warranty. Permissible amplitude criteria from the supplier should override criteria from other sources. If supplier or client criteria are not available, then velocity-based criteria from PIP STC01015 can be used as acceptable limits of motion at rotor bearing points as follows:

rotating machines: 3.0 mm/s

reciprocating machines: 3.8 mm/s

Alternatively, the six parts of BS ISO 10816 provide detailed amplitude criteria.

Note that current Fluor Guideline 1233/1234/1235 criterion is very conservative at lower frequencies with respect to published industry criteria.

Note that the quality of rotor balance is often significantly better than the guideline criteria.

Human Limits: These are used to provide tolerable working conditions for access,

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monitoring, and maintenance. The Reiher-Meister chart (shown below) is commonly used to evaluate human tolerance. Unless stipulated otherwise, the limit should be the line between "easily noticeable to persons" and "troublesome to persons".

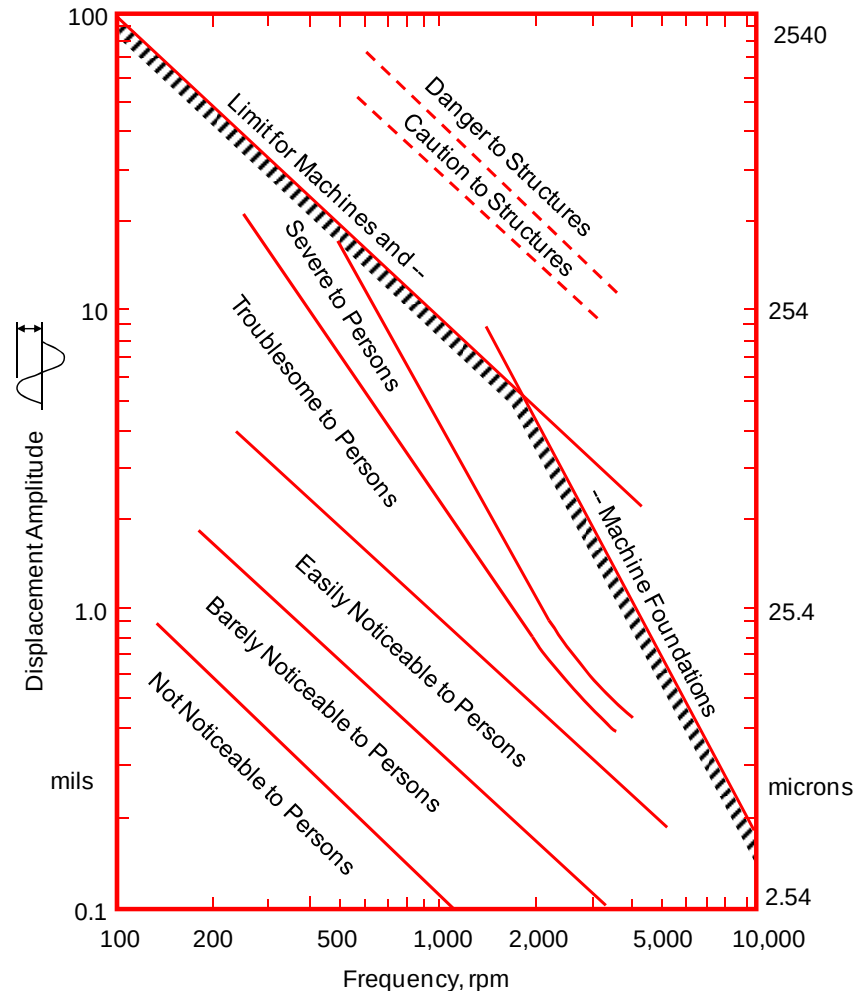


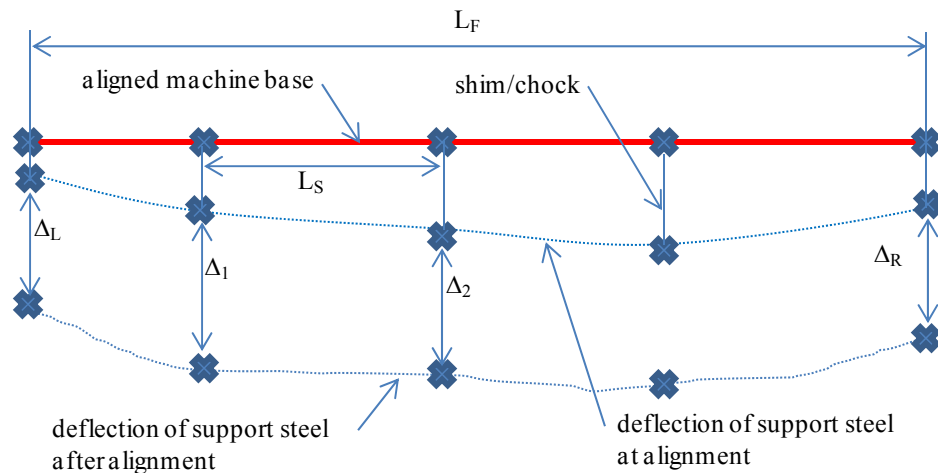
Figure: Reiher-Meister Chart

Superstructure: A machine vibration analysis looks at vibration on the machine for machine tolerance and platforms near the machine for human tolerance. Specifications for concrete supported machines typically state not to tie the adjacent superstructure to the vibrating equipment foundation. Offshore structures have no option to physically separate one from the other. Criterion to evaluate amplitudes and resonance within a superstructure (remote from the machine) is not standardized and is up to the design engineer's judgment.

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Support Rigidity Criteria

Shaft alignment needs to be maintained for optimal machine performance. This implies that the static differential deflection directly beneath the machine must be adequately limited. For typical concrete block foundations, this is not an issue. This could be an issue for a concrete tabletop, however the top slab is usually conservatively sized by judgment. For steel frames directly under a machine, rigidity can be a significant issue, especially if the motor and compressor are on separate skids. The definition of frame distortion, using unfactored loads, is as follows:



For the end points, distortion = $\Delta_L - \Delta_R$

For one or more interior points, distortion = $\Delta_2 - \Delta_1$

Frame distortion applies in either the longitudinal or lateral directions; however module-mounted machines are usually supported only at the two end points.

In lieu of supplier-provided criteria, the distortion between the end points should be limited to $L_F / 2000$. For interior points, the distortion should be limited to $L_S / 5000$. Though not stated, it is presumed that these limits consider final alignment tolerance levels. Low probability high magnitude live loads such as seismic and blast should not be considered. Supplier requirements may include limiting criteria for specific load cases such as a turbine upset.

Support Strength Criteria

Machine dynamic forces are often applied to check the static strength of the supporting structure. The engineer must decide what forces to apply statically. The peak dynamic forces could be applied statically. Alternatively, the peak dynamic forces could be applied after multiplying by a factor to account for dynamic magnification. Another rule of thumb is to apply, in some combination, 50% of the machine weight downward, 25% laterally (transverse to the shaft) and 25% longitudinally.



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Machines are not always as rigid as normally assumed. This could become an issue for larger reciprocating machines where less than ideal machine rigidity can lead to gas pressure loads being transmitted into the supporting frame. Refer to ACI 351.3R-04, Foundations for Dynamic Equipment, Section 3.2.3.2. Note that this is a local effect near the base of the machine because gas pressure forces will cancel each other.

Soil/Pile Load

For traditional isolated supports for vibrating equipment, soil pressure and pile loads are limited in order to ensure predictable linear impedance. For modules, a single structure is provided for the machine and all surrounding facilities. Thus the limiting of soil/pile loading might be prohibitively expensive. Additional geotechnical analysis may then be required to determine appropriate soil/pile impedance values.

ANALYTICAL MODEL

Preparation of the analytical dynamic model requires consideration of a number of issues. One cannot just use the model from the static analysis and add mass and dynamic forces.

Level of Detail

The objective is to prepare a model that is comprehensive enough to behave dynamically as the actual structure, yet economical enough to minimize analysis time and manhours. Typically primary beams, columns, diaphragms, and braces are modeled. Miscellaneous items such as girts, purlins, and architectural elements are modeled only as mass.

Base Frame - This is where the machine is mounted and deserves the most accurate modeling. Diaphragm effects from floor plate need to be included. The machine base, floor plate, and support framing should be modeled in dimensionally correct locations. This implies that standard centerline locations should be adjusted accordingly. SAP2000 includes several means to assist in this effort, such as top of steel options, rigid links, and rendering.

Superstructure - These members are more remote from the machine and somewhat less modeling accuracy is implied. Thus members may be modeled on nominal centerlines.

Often simplified models are prepared from judgment alone. It is suggested to validate models that involve significant differences with the full structure.

Machine - Machines are often assumed by structural engineers to be perfectly rigid with respect to the supporting structure. In contrast, mechanical engineers make the opposite assumption when performing a rotor dynamic analysis. A rigid machine assumption might be reasonable for small pumps on a single skid, however larger equipment depends more on the base frame and supporting structure to maintain shaft alignment between a pump/compressor, gear box, and driver. The supplier's rotor dynamic analysis may indicate how the machine is expected to behave dynamically. The dynamic model used should be that which reasonably mimics the vibration of the actual machine. Because of the configuration of the bearings, the shaft itself does not provide a rigid axial link between machine components. The following is suggested as a minimum:



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- Model joints at the location of mass CGs, location of amplitude measurement points, location where dynamic loads are applied, and locations where the machine components are connected to the base frame and/or supporting structure.
- Model pump/compressor, gear box, and driver as separate rigid components using either rigid link elements or body constraints.
- Model the supplier's base frame if it is judged to be significant relative to the supporting frame stiffness.

Stiffness

Framing Member stiffness is automatically calculated by SAP2000 based on frame member properties.

Vibration Isolators can be modeled using link (2 joint) elements. Refer to Attachment 4, Listing 3.

Soil/Piles without damping can be modeled using spring properties. Refer to Attachment 4, Listing 1.

Soil/Piles with damping can be modeled using support (1 node) elements. Refer to Attachment 4, Listing 2.

Soil/Piles with frequency dependent impedance can be modeled using frequency dependent properties assigned to a support element. Refer to Attachment 4, Listing 4. Note that frequency dependent properties only apply to a Steady-State analysis.

Member property modifications can be applied if judged to be necessary. Modifications might be due to the difference between theoretical fixed/pinned connection details and actual behavior, or due to potential slippage of snug tight bolted connections. For now, this is up to the engineer's discretion. Refer to Attachment 4, Listing 5.

Material Damping

This is the damping resulting from interparticle friction. If a single value for the entire model is desired, a value may be entered in the load case dialog box. Refer to Attachment 4, Listing 6.

If separate values of damping are desired for each material, values may be added in the material dialog box. To enter material-specific damping, refer to Attachment 4, Listing 7.

Damping values from the load case and material represent the same phenomena and are additive to each other. The damping value entered into SAP2000, depends on the type of dynamic analysis to be run:

modal time-history: The program is looking for a damping ratio ($\varepsilon = C / C_{CR}$). For a damping ratio of 5%, enter 0.05.



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steady-state: The program is looking for hysteretic proportional damping coefficients ($D = C \omega = \alpha M + \beta K$). Enter $\alpha = 0$ and $\beta = 2 \varepsilon$. For a damping ratio of 5%, enter $\beta = 0.10$.

Discrete Damping

This applies to specific damping values (i.e. dashpots) applied at user-specified locations.

Vibration Isolators: Damping can be modeled using a link (2 joint) element. Constant values of stiffness and damping can be entered using the data screens for these elements. Refer to Attachment 4, Listing 3.

Soil/Piles: Damping can be modeled using a support (1 joint) element. Constant values of stiffness and damping can be entered using the data screens for these elements. Refer to Attachment 4, Listing 2.

Frequency Dependent: Damping can be entered using a frequency dependent properties object. This object is then assigned to the applicable link or support elements. Refer to Attachment 4, Listing 4.

Modal Time-History: This type of analysis will obtain damping values from frequency independent data (frequency dependent properties, if provided, are ignored). The damping obtained is described as *effective* damping coefficient in units of force-seconds/length. Input cross damping values are ignored.

Steady-State: This type of analysis will obtain damping values from the frequency dependent data, if provided, with damping interpreted as a *hysteretic* damping coefficient (D) in units of force/length. Otherwise impedance values are taken from frequency independent data with damping interpreted as a *viscous* damping coefficient (C) in units of force-seconds/length.

Dyna5 & DynaN damping results are *viscous* (C). The relationship between viscous & hysteretic damping is $D = C \Omega$.

Mass

Member Mass: This is the mass of each modeled element. This mass is calculated automatically by SAP2000 based on material mass density. To assign mass density to a material, refer to Attachment 4, Listing 10. Member mass can also be assigned through a converted load pattern. To add mass through a converted load pattern, refer to Attachment 4, Listing 9.

Discrete Mass: This is mass that is in addition to member mass. Discrete mass is applied at a joint through direct assignment or through a converted load case. The advantage of converting a load case is that that load magnitudes can be displayed on screen. To directly assign mass to a joint, refer to Attachment 4, Listing 8. To add mass through a converted load pattern, refer to Attachment 4, Listing 9.



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Element Size

Element size criterion mainly affects high speed machines. To assure reasonable stress wave propagation through a structure, the maximum element size is established as one tenth of a single cycle of the machine harmonic force. A detailed discussion can be found in "Finite Element Procedures", Klaus-Jurgen Bathe, Prentice Hall, 1996. A simple example follows:

machine speed, $\Omega = 7,500 \text{ rpm} = 125 \text{ Hz}$

period, $T = 1 / \Omega = 1 / 125 \text{ Hz} = 0.008 \text{ s}$

material: steel, shear wave velocity, $V_s = 3,130 \text{ m/s}$

length of wave, $L_w = (V_s)(T) = (3,130 \text{ m/s})(0.008 \text{ s}) = 25 \text{ m}$

maximum element length, $L_E = L_w / 10 = (25 \text{ m}) / 10 = 2.5 \text{ m}$

For steel structures originally modeled for a static analysis, the element spacing could easily exceed 2.5 m. Meshing is a quick, change-tolerant, means to allow for wave propagation. Refer to Attachment 4, Listing 11.

Load Phasing

Amplitudes resulting from machine and driver dynamic loads, or from multiple machines in the same structure, create an issue on to how to calculate and combine amplitudes in order to compare with permissible values which are often speed-based.

Machine and driver can always be analyzed separately with the results manually added together as appropriate. This approach will be necessary if the speeds are different and a steady-state analysis is being used. If a time-history analysis is used, machine and driver forces can be combined in a single load case. In this case the phasing of loads between machine and driver will then need to be considered.

When motor and machine are rotating in the same direction, in-phase and 180 degree out of phase combined cases are typically considered. These two cases are significant only if the machine and driver frequencies are the same, or an even multiple of each other. If not at an even multiple, the phasing of loads will not stay the same and a much longer duration would be needed to capture maximum amplitudes.

Note that with damping present, resulting peak amplitudes will not be in-phase with the peak applied load. Thus combined load cases will usually lead to lower amplitudes, as opposed to the simple addition of separately computed peak amplitudes.

Given a rotating shaft imbalance creating a harmonic dynamic load, the direction of shaft rotation (i.e. clockwise or counterclockwise) will produce different amplitudes. This is a damping-induced effect. With a greater magnitude of damping, this difference will be greater.



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If a gear box is used, the machine & driver rotation likely will be in opposite directions. A single modal time-history load case can still be run, provided that vertical and horizontal loads are input properly.

Thermal Sliding Supports

For large modules, sliding supports are often used to reduce thermal expansion resulting from large temperature differences between summer and winter. For a vibratory analysis, induced stresses are usually small enough that sliding will not result. Thus these supports can be considered restrained if the static load cases indicate that the support will always have adequate compression.

DYNAMIC ANALYSIS:

A dynamic analysis is performed in order to determine natural frequencies and amplitudes. SAP2000 has several options on how this can be performed, however a direct integration time-history not considered in this guideline due to the lengthy analysis time usually required.

A static analysis is performed to evaluate frame deformations for pre and post alignment situations.

Modal Transformation

A modal transformation is commonly used to determine undamped natural frequencies. Undamped natural frequency values and mass participation are reported by SAP2000 as part of the modal analysis results. There is some debate about the need to use damped or undamped natural frequencies. If desired, damped natural frequencies could be manually calculated from SAP2000 modal damping results (Attachment 4, Listing 13). SAP2000 results show mass participation for every mode calculated in the analysis. However the results table can be filtered to show only modes where the mass participation is greater than (for example) 5%.

Number of Modes - The generic approach is to run the analysis model using an increasing numbers of modes in order to determine where resulting amplitudes stop changing. For seismic designs, 90% mass participation is used; however this criterion may not produce an accurate local forced response result. It is recommended that the number of modes analyzed results in a minimum of 90% mass participation *and* includes frequencies that are at least 120% of the highest machine frequency.

Ritz Vectors vs. Eigen Vectors - Ritz vectors are load-dependent and mobilize mass faster than Eigen vectors, thus producing accurate results faster and using fewer modes. For a frequency analysis, the first 10 or so modes will be the same. For higher modes, Ritz frequencies will be higher than Eigen frequencies. Thus it is recommended to use the Eigen option for a frequency analysis unless specific agreement is received from the client.



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Modal Time-History Analysis

This is a time-domain numerical integration of the equation(s) of motion. SAP2000 provides several implicit methods of advancing deflection, velocity, and acceleration to the next time step, though so far no significant result differences have been observed. The time step, steps per cycle, number of cycles, and output steps are significant in obtaining an accurate result. These input variables are provided in the Time History Function and Load Case as follows:

<u>Time History Function</u>	<u>Load Case</u>
period	transient or periodic
steps/cycle	output steps
cycles	output time step

Variables derived from input values are as follows:

<u>Time History Function</u>	<u>Load Case</u>
increment = (period)/(steps/cycle)	increment = output time step
duration = (period)(cycles)	duration = (steps)(increment)

Increment: The increment should be no greater than one tenth of a load cycle, *and* one tenth of the highest mode period.

Duration: The time-history duration that SAP2000 uses is from the Load Case.

Steps: The number of time-history steps that SAP2000 uses is computed from both the Time History Function and Load Case. To minimize the analysis time, the Time History Function and Load Case increments should be equal, or one should be an integer multiple of the other.

Output: If needed to minimize output, the Load Case increment should be a higher integer multiple of the Load Definition increment.

Periodic Option: This option attempts to automatically remove the transient portion of the response for periodic (not necessarily harmonic) loads. For this to work properly, the time-history duration must be an integer multiple of the load cycle duration. For the case of a motor and compressor operating at different speeds, the time-history duration must be an integer multiple of both load cycles. An example follows:

motor speed: 9581 rpm
compressor speed: 11,871 rpm

Unless these two speeds can be reduced by a common factor, then the duration must be set to 60 seconds (i.e. 9581 motor cycles & 11,871 compressor cycles)

Steady-State Analysis

This is a closed form frequency-domain solution of the equation(s) of motion based on a harmonic forcing function. The method is often used for rigid block foundations with 1,



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2, or 6 coupled degrees of freedom (i.e. Dyna5, DynaN). For a typical situation with a machine and driver operating at different speeds, two steady-state cases would be needed (one for each operating speed). The resulting amplitudes are complex (in the form of $A = a + ib$) and can then be converted to a magnitude (Δ) and phase angle (ϕ) in degrees as follows:

$$\Delta = \text{modulus } (A) = \sqrt{a^2 + b^2} \quad (5a)$$

$$\phi = \text{argument } (A) \quad (5b)$$

An argument function is included in Excel and MathCad, however it can be manually evaluated from the values of a and b as follows:

	$a < 0$	$a = 0$	$a > 0$
$b < 0$	$-180 + \arctan(b/a)$	-90	$\arctan(b/a)$
$b = 0$	-180	(no amplitude)	0
$b > 0$	$+180 + \arctan(b/a)$	$+90$	$\arctan(b/a)$

Analysis time for a single steady-state run can be dramatically faster than an equivalent modal time-history. However, the analysis time for a frequency range of steady-state runs might not be faster than a modal separation.

SAP2000 can perform a coupled steady-state analysis, but not a modal steady-state analysis. However a steady-state analysis can evaluate at natural frequencies computed from a modal transformation. The program can handle phased vertical and horizontal load components (with a common operating speed) in a single load case. Results are presented as real and imaginary components, or as magnitudes. If phase angle results are desired, the equation presented above can be used. Refer to Attachment 4, Listing 19.

A steady-state analysis could also be used to detect damped or undamped natural frequencies. This form of analysis can also calculate amplitudes over a range of frequencies. Unlike a modal transformation, a steady-state range does not need to start at a frequency of zero. If the specified resonant range is 0.8 to 1.2 times machine speed, then the analysis can be run (with some margin) from 0.75 to 1.25 times machine speed. Evaluation of coastdown conditions would require a wider range. Natural frequencies become detectable as peaks in an amplitude-frequency graph, provided that damping is sufficiently suppressed from the model. The use of a steady-state analysis for this purpose is common for block-type foundations (i.e. Dyna5, DynaN) and is ideally suited for the evaluation of frequency dependent soil/pile impedance. For a steel frame structure, the resonant peaks will only partially correlate to high modal mass participation frequencies. This is because modal mass participation corresponds to movement of the entire structure, while steady-state amplitudes correspond to movement only at the selected measurement location. If a location on the machine is selected, then the peak amplitudes would indicate which natural frequencies directly affect the machine.

The advantages to using a steady-state analysis are:

1. Analysis at a single frequency is faster than modal time-history, even if the lengthy



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modal transformation portion is excluded.

2. There is a more accurate handling of damping. A modal time-history uses an approximate means to apportion damping to the individual modes.
3. There is no need to deal with the number of modes, time step, duration, or time-history results that might exceed disk capacity.

EVALUATION OF RESULTS

Frequency Analysis

Modal analysis results can be retrieved through animation and tables (refer to Attachment 4, Listings 14 and 15).

If frequency criterion is not met, then a revision of the model's mass and/or stiffness will become necessary. This will involve a resizing of the structural configuration or element sizing.

If frequency criterion cannot reasonably be met, then an agreement of the acceptability of modes near the machine speed should be pursued with the client and (if applicable) the supplier. One approach is to use 50% of the damping normally applied, and show that computed amplitudes remain within acceptable limits with the forcing function speed matching the natural frequency that is within the resonant range.

Forced Response Analysis

Forced response results can be retrieved through animation, graphs, and tables (Refer to Attachment 4, Listings 16 through 19).

If response criterion is not met, then a revision of the model's mass and/or stiffness will become necessary. This will involve revising the structural configuration, foundation support points, or element sizing.

ADDITIONAL TOPICS

Anchorage & Leveling

Connecting a machine to an elevated concrete or steel support is as important as for grade-level block foundations. Lateral forces are transmitted through friction, thus the purpose of bolts is to remain in tension and produce compression between the machine base and support structure. Proprietary bolts designed for leveling and/or tension control can be used.

Mounting may be through the use of shims or grout (chocks). With either mounting method, support is provided at discrete locations, not continuously. The mechanical engineer will need to be consulted on the mounting method and the gap needed for shims or grout.



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Transport Disassembly

Disconnecting machine components for module transport is not a structural decision. If need be, input from the mechanical engineer on disassembly can be requested. If the machine will not have shaft elements decoupled and piping disconnected, the mechanical and pipe stress criteria for acceptable deflection of the structure will be a critical part of the analysis. The required structure stiffness evaluated based on its support during assembly and all the steps in handling through to final setting may govern the design. This check should be performed before the dynamic analysis.

NOMENCLATURE

a	real component of a complex number
b	imaginary component of a complex number
C	viscous damping coefficient (kN-s/mm)
C_{CR}	critical viscous damping coefficient (kN-s/mm)
D	hysteretic damping coefficient (kN/mm)
E	elastic modulus (MPa)
E_{dyn}	dynamic modulus (MPa)
K	stiffness
M	mass
L_W	length of wave (m)
L_E	maximum element length (m)
T	period (s)
V_S	shear wave velocity (m/s)
α	proportional damping mass coefficient
β	proportional damping stiffness coefficient
ε	damping ratio
Ω	machine frequency (Hz)
ω	natural structure frequency (rad/s)
Δ	deflection magnitude (m)
ϕ	phase angle (rad)

REFERENCES

API RP 686 (PIP REIE 686), Recommended Practice for Machinery Installation and Installation Design, American Petroleum Institute, Washington, DC

BS ISO 10816, Mechanical Vibration - Evaluation of Machine Vibration by Measurements on Non-Rotating Parts, British Standards Institution, London, UK

BS ISO 1940, Mechanical Vibration - Balance Quality Requirements for Rotors in a Constant (Rigid) State, British Standards Institution, London, UK

ACI 351.3R, Foundations for Dynamic Equipment, American Concrete Institute, Farmington Hills, MI

NON-RIGID SUPPORT FOR VIBRATORY EQUIPMENT

SAP2000, Computers & Structures, Inc., Berkeley, CA

Dyna5, University of Western Ontario, London, ON

DynaN, Ensoft Inc., Austin, TX

Civil/Structural Engineering:

[Guideline 000.215.1233](#) Vibrating Machinery Foundations on Soil (Frequency Independent Criteria)

[Guideline 000.215.1234](#) Vibrating Machinery Foundations on Soil

[Guideline 000.215.1235](#) Vibrating Machinery Foundations on Piles

[Specification 000.215.00920](#) Structural Data for Mechanical Equipment

[Specification 000.215.STC01015](#) Structural Design Criteria

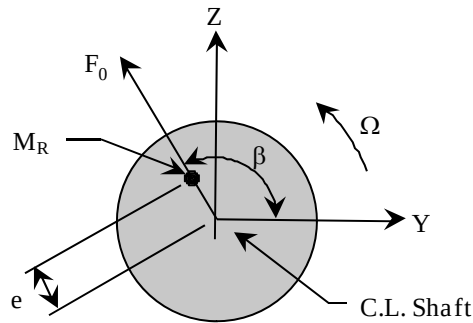
[Specification 000.210.CVS02010](#) Geotechnical Engineering Investigation Specification

ATTACHMENTS

Attachment 01	Rotating Machine Forces
Attachment 02	Reciprocating Machine Forces
Attachment 03	Foundation Impedance
Attachment 04	SAP2000 Implementation Notes

NON-RIGID SUPPORT FOR VIBRATORY EQUIPMENT**Rotating Machine Forces**

Inertia forces are caused by imperfect balancing of a rotor due to manufacture or due to usage. Forces from this imbalance are calculated as follows:



$$F_0 = (M_R)(e)(\Omega)^2 / 1000 \quad (A1-1)$$

$$F_X(t) = 0 \quad (A1-2a)$$

$$F_Y(t) = F_0 \cos [(\Omega)(t) + \beta] \quad (A1-2b)$$

$$F_Z(t) = F_0 \sin [(\Omega)(t) + \beta] \quad (A1-2c)$$

where,

- F_0 = inertia force of unbalanced rotor (N)
- M_R = rotor mass (kg)
- e = eccentricity of rotor (mm)
- Ω = machine frequency (rad/s)
- t = time (s)
- F_X, F_Y, F_Z = forces in each coordinate direction (N)
- β = crank angle (rad)

The machine weight and frequency are always supplied. The crank angle (β) becomes relevant when adding the effects of more than one machine, such as a pump and motor. The design engineer will then select the angle that produces the worst case result. In contrast, data to compute the unbalanced force may not always be supplied.

The preferred choice is to obtain the unbalanced force (F_0) directly from the machine supplier. This is nearly always supplied for the largest machines, but rarely supplied for small machines.

The second choice is to check with Mechanical to identify the quality grade of the rotor. The quality grade is associated with a value (Q_G) representing eccentricity multiplied by frequency. Values of the quality grade and associated Q_G are shown in Table A1. This, along with the rotor mass enables the unbalanced force as

$$F_0 = (M_R)(Q_G)(\Omega) S_F / 1000 \quad (A1-3)$$

where,

- Q_G = eccentricity times the machine frequency, (m/s)
- S_F = service factor to allow for long-term degradation, usually 2 or more



NON-RIGID SUPPORT FOR VIBRATORY EQUIPMENT

Rotating Machine Forces

Note that vendor quality values may be intended to be used without degradation to determine acceptable amplitudes. It would be better understood that the machine balance would initially be better than the required minimum quality, and the acceptable vibration would be measure at start of operation.

If the quality grade is not identified, the initial test eccentricity of the rotor can be used along with Equation A1-1. Vibration sensors may also be examined to give an indication about an upper limit of shaft misalignment. If such sensors are used, then an eccentricity may be computed based on the shaft vibration level that will cause the machine to be shut down. Judgment must be applied on an individual case basis with the assistance of the Mechanical Engineer.

The next choice, if eccentricity data are not available, is to use a common rule-of-thumb formula derived from Equation A1-3 and based on a Q_G value of 16 mm/s

$$\text{force, kN} = \frac{(\text{rotor weight, kN})(\text{rotor speed, rpm})}{6,000} \quad (\text{A1-4})$$

The last choice is if the rotor weight is not available. In this case, an approximate rotor weight can be estimated based on the ratio of rotor weight to machine weight for similar equipment. Then Equation A1-4 can be applied.

Whichever formula is selected in lieu of supplier information, the design criteria should be reviewed with the machine supplier for acceptance.

Table A1: Rotor Quality Grade and Q_G (from BS ISO 1940)

Machines	Grade	Q_G (mm/s)
Crushing Machines	G 16	16
General Machinery, fans, gears, pumps	G 6.3	6.3
Compressors, gas/steam turbines	G 2.5	2.5
Grinding machine drives	G 1	1.0

Note that the quality grade for the machine being purchased may not correspond to the generic descriptions in this table. If is possible that pump rotors will be G 2.5 or G1. Also, the motor rotor quality may be better than the pump rotor. This can make a significant difference to the resulting vibration amplitudes.

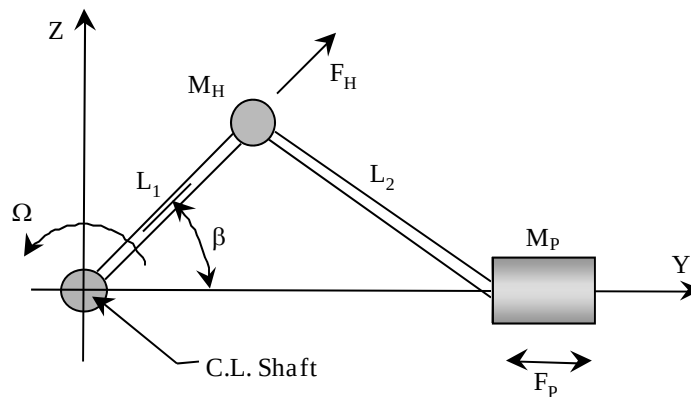
NON-RIGID SUPPORT FOR VIBRATORY EQUIPMENT

Reciprocating Machine Forces

Two types of forces act within a reciprocating compressor: inertial forces and gas forces.

For the dynamic analysis of a mat, gas forces tend to cancel each other within the machine frame and within the foundation pier. Thus only inertia forces need be considered. However, for machine anchorage and pier design purposes, the magnitude of gas forces transmitted into the pier is very important and is dependent on machine rigidity. To obtain reasonably accurate anchorage design forces, the rigidity of the machine must be realistically considered.

Inertia forces result from varying accelerations of rotating and reciprocating machine parts and act at the crankshaft bearings without any opposing reaction on the frame. The magnitude of inertia forces depend on speed, rotating inertia, and on reciprocating inertias of crosshead, piston rod, and piston as follows:



For one cylinder:

$$F_X(t) = 0 \quad (A2-1a)$$

$$F_Y(t) = (M_H + M_P)(L_1)(\Omega)^2 \cos [(\Omega)(t) + \beta] + (M_P / L_2)(L_1 \Omega)^2 \cos 2 [(\Omega)(t) + \beta] \quad (A2-1b)$$

$$F_Z(t) = (M_H)(L_1)(\Omega)^2 \sin [(\Omega)(t) + \beta] \quad (A2-1c)$$

Where:

- M_H = hinge mass (kg)
- M_P = piston mass (kg)
- L_1 = length from shaft to hinge (m)
- L_2 = length from hinge to piston (m)
- Ω = machine speed (rad/s)
- t = time (s)
- F_X, F_Y, F_Z = forces in each coordinate direction (N)
- F_P = inertia force from piston (N)
- F_H = inertia force from hinge (N)
- β = crank angle (rad)

The last term in Equation A2-1b results in a harmonic force operating at twice the machine frequency. This is the secondary force that is a part of reciprocating machine design.



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Reciprocating Machine Forces

For multiple cylinders, the individual piston forces are combined using the appropriate distances between cylinders and the appropriate crank angles. These combined forces are usually provided midway between the cylinders by the machine supplier.

Gas forces result from the action of piston motion and valve action which generate time varying head and crank end pressures. The magnitude of gas forces depend on the differential between suction and discharge pressure, the area of the cylinder bore, rod area, pulsations, and external resistance. The near uniform pressure within each compressed volume make the force on the cylinder a direct reaction to the force on the piston. Thus, gas forces act on the crankshaft with an equal and opposite reaction on the cylinder. Refer to ACI 351.3R for a more detailed description and equations.



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Foundation Impedance

Impedance from soil or piles is not quite as significant as for rigid block foundations, but some rational and documented value should be included. Variations should also be considered for field measurement accuracy, and summer/winter/spring thawing conditions as applicable.

Soil or pile impedance can be obtained or determined by several methods based on measured soil properties. The following lists these methods in order of preference:

- 1) from a geotechnical engineer (who is experienced in this type of design)
- 2) from commercial software (i.e. Dyna5, DynaN)
- 3) manually calculated from Fluor Guidelines (1233, 1234, 1235) or from a recognized published reference.

Converting Dyna/DynaN Group Impedance Results into SAP2000 Input

Impedance results obtained from Dyna5/DynaN are reported in terms of group stiffness and damping values. A finite element SAP2000 model normally includes a series of joints at the bottom of the mat or pile cap. For pile supported structures, each joint represents a pile location. For soil supported structures, an array of joints is used to represent the distribution of soil pressure.

The purpose of this section is to describe a method to translate group impedance into equivalent discrete impedance values. Actual designs tend to use a variety of procedures; many tend to overlook several of the group values. Most commonly, the three translation stiffness/damping values are divided by the number of piles while ignoring all rotational and cross values.

The following procedure assumes a rigid mat/pile cap and that all mat/pile cap joints will have the same individual impedance properties. This second assumption greatly simplifies input into SAP2000.

Variables:

k	single pile stiffness
K	group pile stiffness
N	number of piles
dist	distance from a pile to the center of the pile pattern, perpendicular to the subscript axis
c	single pile viscous damping
C	group pile viscous damping

Subscripts:

X	horizontal
Y	horizontal
Z	vertical
C	cross

Equations For stiffness,

$$\begin{aligned} k_X &= K_X / N & (A3-1a) \\ k_Y &= K_Y / N & (A3-1b) \\ k_Z &= K_Z / N & (A3-1c) \end{aligned}$$

NON-RIGID SUPPORT FOR VIBRATORY EQUIPMENT

Foundation Impedance

$$k_{RX} = [K_{RX} - k_v (\sum \text{dist}^2)_X] / N \quad (\text{A3-2a})$$

$$k_{RY} = [K_{RY} - k_v (\sum \text{dist}^2)_Y] / N \quad (\text{A3-2b})$$

$$k_T = [K_T - k_X (\sum \text{dist}^2)_X - k_Y (\sum \text{dist}^2)_Y] / N \quad (\text{A3-2c})$$

$$k_{CX} = K_{CX} / N \quad (\text{A3-3a})$$

$$k_{CY} = K_{CY} / N \quad (\text{A3-3b})$$

Equations For damping,

$$c_X = C_X / N \quad (\text{A3-4a})$$

$$c_Y = C_Y / N \quad (\text{A3-4b})$$

$$c_Z = C_Z / N \quad (\text{A3-4c})$$

$$c_{RX} = [C_{RX} - c_v (\sum \text{dist}^2)_X] / N \quad (\text{A3-5a})$$

$$c_{RY} = [C_{RY} - c_v (\sum \text{dist}^2)_Y] / N \quad (\text{A3-5b})$$

$$c_{RZ} = [C_{RZ} - c_X (\sum \text{dist}^2)_X - c_Y (\sum \text{dist}^2)_Y] / N \quad (\text{A3-5c})$$

$$c_{CX} = C_{CX} / N \quad (\text{A3-6a})$$

$$c_{CY} = C_{CY} / N \quad (\text{A3-6b})$$

Example: Distribution of Pile Impedance on a FEM Mat

The pile pattern from Figure A3-1 is first run in Dyna5/DynaN in order to obtain group impedance values. The pile pattern orientation should be arranged such that the X axis in Dyna5 is parallel to the X axis in SAP2000. In addition, dimensional parameters, "distance from base center to CG", should be set to zero. Dyna5 or DynaN will report impedance at the CG as shown in Figure A3-2.

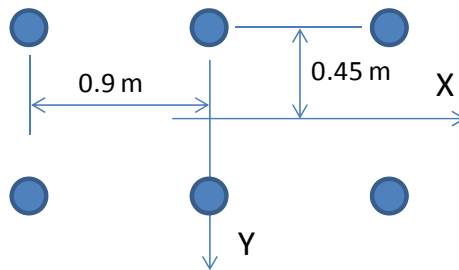


Figure A3-1: Pile Pattern

NON-RIGID SUPPORT FOR VIBRATORY EQUIPMENT

Foundation Impedance

```

FREQUENCY -- 25.000 Hertz
*****

STIFFNESS CONSTANTS (K)
*****

HORIZONTAL TRANSLATION (X) ... 1.218E+08 N/m
HORIZONTAL TRANSLATION (Y) ... 1.307E+08 N/m
VERTICAL TRANSLATION (Z) ..... 8.288E+08 N/m
ROTATION ABOUT (X) ..... 4.749E+08 N.m/rad
ROTATION ABOUT (Y) ..... 8.863E+08 N.m/rad
TORSION ABOUT (Z) ..... 1.417E+08 N.m/rad
CROSS-STIFFNESS (YZ PLANE) .... 9.413E+07 N/rad
CROSS-STIFFNESS (XZ PLANE) .... -9.011E+07 N/rad

DAMPING CONSTANTS (C)
*****

HORIZONTAL TRANSLATION (X) ... 7.293E+05 N/m/s
HORIZONTAL TRANSLATION (Y) ... 7.465E+05 N/m/s
VERTICAL TRANSLATION (Z) ..... 4.482E+06 N/m/s
ROTATION ABOUT (X) ..... 2.732E+05 N.m/rad/s
ROTATION ABOUT (Y) ..... 6.713E+05 N.m/rad/s
TORSION ABOUT (Z) ..... 3.061E+05 N.m/rad/s
CROSS-DAMPING (YZ PLANE) ..... 3.876E+05 N/rad/s
CROSS-DAMPING (XZ PLANE) ..... -3.900E+05 N/rad/s

```

Figure A3-2: Dyna5 output

The results shown in Figure A3-2 are interpreted as follows,

$K_X = 121,800 \text{ kN/m}$	$C_X = 729.3 \text{ kN-s/m}$
$K_Y = 130,700 \text{ kN/m}$	$C_Y = 746.5 \text{ kN-s/m}$
$K_Z = 828,800 \text{ kN/m}$	$C_Z = 4,482 \text{ kN-s/m}$
$K_{RX} = 474,900 \text{ kN-m/rad}$	$C_{RX} = 273.2 \text{ kN-m-s/rad}$
$K_{RY} = 886,300 \text{ kN-m/rad}$	$C_{RY} = 671.3 \text{ kN-m-s/rad}$
$K_{RZ} = 141,700 \text{ kN-m/rad}$	$C_{RZ} = 306.1 \text{ kN-m-s/rad}$
$K_{CX} = -90,110 \text{ kN/rad}$	$C_{CX} = -390.0 \text{ kN-s/rad}$
$K_{CY} = +94,130 \text{ kN/rad}$	$C_{CY} = +387.6 \text{ kN/rad}$

With the pile layout from Figure A3-1,

number of piles, $N = 6$

pattern parameters,

$$(\Sigma \text{dist}^2)_X = (6 \text{ each})(0.45 \text{ m})^2 = 1.215 \text{ m}^2$$

$$(\Sigma \text{dist}^2)_Y = (4 \text{ each})(0.9 \text{ m})^2 = 3.24 \text{ m}^2$$

single pile translation stiffness using equation A3-1,

$$k_X = K_X / N = (121,800 \text{ kN/m}) / (6 \text{ each}) = 20,300 \text{ kN/m}$$



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Foundation Impedance

$$k_Y = K_Y / N = (130,700 \text{ kN/m}) / (6 \text{ each}) = 21,783 \text{ kN/m}$$

$$k_Z = K_Z / N = (828,800 \text{ kN/m}) / (6 \text{ each}) = 138,133 \text{ kN/m}$$

single pile rocking stiffness using equation A3-2,

$$k_{RX} = [K_{RX} - k_Z (\Sigma \text{dist}^2)_X] / N \\ = [(474,900 \text{ kN-m/rad}) - (138,133 \text{ kN/m})(1.215 \text{ m}^2)] / (6 \text{ each}) = 51,178 \text{ kN-m/rad}$$

$$k_{RY} = [K_{RY} - k_Z (\Sigma \text{dist}^2)_Y] / N \\ = [(886,300 \text{ kN-m/rad}) - (138,133 \text{ kN/m})(3.24 \text{ m}^2)] / (6 \text{ each}) = 73,125 \text{ kN-m/rad}$$

$$k_{RZ} = [K_{RZ} - k_X (\Sigma \text{dist}^2)_X - k_Y (\Sigma \text{dist}^2)_Y] / N \\ = [(141,700 \text{ kN-m/rad}) - (20,300 \text{ kN/m})(3.24 \text{ m}^2) - (21,783 \text{ kN/m})(1.215 \text{ m}^2)] / (6 \text{ each}) = 8,244 \text{ kN-m/rad}$$

single pile cross stiffness using equation A3-3,

$$k_{CX} = K_{CX} / N = (-90,110 \text{ kN/rad}) / (6 \text{ each}) = -15,018 \text{ k/rad}$$

$$k_{CY} = K_{CY} / N = (+94,130 \text{ kN/rad}) / (6 \text{ each}) = +15,688 \text{ k/rad}$$

single pile translation damping using Equation A3-4,

$$c_X = C_X / N = (729.3 \text{ kN-s/m}) / (6 \text{ each}) = 121.6 \text{ kN-s/m}$$

$$c_Y = C_Y / N = (746.5 \text{ kN-s/m}) / (6 \text{ each}) = 124.4 \text{ kN-s/m}$$

$$c_Z = C_Z / N = (4,482 \text{ kN-s/m}) / (6 \text{ each}) = 747 \text{ kN-s/m}$$

single pile rocking damping using Equation A3-5,

$$c_{RX} = [C_{RX} - c_Z (\Sigma \text{dist}^2)_X] / N \\ = [(273.2 \text{ kN-m-s/rad}) - (747 \text{ kN-s/m})(1.215 \text{ m}^2)] / (6 \text{ each}) = -105.7 \text{ kN-m/rad}$$

$$c_{RY} = [C_{RY} - c_Z (\Sigma \text{dist}^2)_Y] / N \\ = [(671.3 \text{ kN-m-s/rad}) - (747 \text{ kN-s/m})(3.24 \text{ m}^2)] / (6 \text{ each}) = -291.5 \text{ kN-m/rad}$$

$$c_{RZ} = [C_{RZ} - c_X (\Sigma \text{dist}^2)_X - c_Y (\Sigma \text{dist}^2)_Y] / N \\ = [(306.1 \text{ kN-m/rad}) - (121.60 \text{ kN/m})(3.24 \text{ m}^2) - (124.4 \text{ kN/m})(1.215 \text{ m}^2)] / (6 \text{ each}) = -39.8 \text{ kN-m/rad}$$

single pile cross damping using Equation A3-6,

$$c_{CX} = C_{CX} / N = (-390.0 \text{ kN-s/rad}) / (6 \text{ each}) = -65.0 \text{ kN-s/rad}$$

$$c_{CY} = C_{CY} / N = (+387.6 \text{ kN-s/rad}) / (6 \text{ each}) = +64.6 \text{ kN-s/rad}$$



NON-RIGID SUPPORT FOR VIBRATORY EQUIPMENT

Foundation Impedance

Within SAP2000, create a (1 joint) support element at each of the six pile locations using the default orientation (refer to Attachment 4, Listing 2). Input the computed stiffness and damping values as shown in Figure A3-3.

Note that local axes orientations are used. Thus the cross stiffness and damping for the U3-R2 (Y translation, X rotation) uses a reversed sign for input into SAP2000.

Linear Link/Support Directional Properties

Link/Support Name: SinglePile

Directional Control:

Direction	Fixed
☒ U1	☐
☒ U2	☐
☒ U3	☐
☒ R1	☐
☒ R2	☐
☒ R3	☐

Shear Distance from End J:

U2: 0.

U3: 0.

Units: KN, m, C

Upper Stiffness Matrix (Symmetrical) Used For All Analysis Cases

☐ Stiffness Is Uncoupled ☒ Stiffness Is Coupled

	U1	U2	U3	R1	R2	R3
U1	138133.	0.	0.	0.	0.	0.
U2		21783.	0.	0.	0.	-15.69
U3			20300.	0.	-15.02	0.
R1				8244.	0.	0.
R2					51178.	0.
R3						73125.

Upper Damping Matrix (Symmetrical) Used For All Analysis Cases

☐ Damping Is Uncoupled ☒ Damping Is Coupled

	U1	U2	U3	R1	R2	R3
U1	747.	0.	0.	0.	0.	0.
U2		124.42	0.	0.	0.	-0.0646
U3			121.55	0.	-0.065	0.
R1				-39.81	0.	0.
R2					-105.73	0.
R3						-291.5

OK Cancel

Figure A3-3: SAP2000 Support Element Input

Alternatively, foundation impedance can be input using frequency dependent properties. In this case, a frequency dependent property would need to be created and assigned to the associated support element (refer to Attachment 4, Listing 4). Values are as computed above, except that damping is input as hysteretic (D), with $D = C \Omega$.

NON-RIGID SUPPORT FOR VIBRATORY EQUIPMENT

SAP2000 Implementation Notes

This attachment is intended to assist in SAP2000 data entry for those with minimal or periodic usage. The following is based on SAP2000, version 15.1.0. The furnished listings are for tasks typical for a dynamic analysis, and consist of the following:

Listing 1: Create a Joint Spring	2
Listing 2: Create a (1 Joint) Support Element	3
Listing 3: Create a (2 Joint) Link Element	5
Listing 4: Define and Assign Frequency Dependent Properties	5
Listing 5: View or Modify Member Stiffness Properties	6
Listing 6: View or Modify Load Case Damping	6
Listing 7: View or Modify Material-Specific Damping	7
Listing 8: Directly Specify and Assign Mass at a Joint	8
Listing 9: Specify that Mass be Computed from a Load Pattern	9
Listing 10: View or Modify Material Mass Density	10
Listing 11: Assign Automatic Meshing to a Frame Member	10
Listing 12: Access Solver Options	11
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General Notes & Suggestions

Numerical Constraints: The numerical accuracy for SAP2000 is 16 digits. Accuracy issues can result at joints where the difference between member stiffness can challenge the 16 digit limit. SAP2000 will generate warning messages when it detects such a discrepancy.

File Location: SAP2000 data files should be saved to the computer's hard drive in order to minimize the time required to load and save files.

Unattended Analysis: Without user involvement, Fluor's computers go into sleep mode after 1 hour. Any SAP2000 analysis running at that time will be involuntarily halted and SAP2000 will likely generate an error message.

Laptop Video Cards: Fluor's standard laptop computer configuration uses a video card that is not aligned with SAP2000 video requirements. Frequent screen pauses during SAP2000 use are apparent. In addition, the program can suddenly crash if the DirectX display option is attempted.

Solver Options: There are three solver options, Regular, Advanced, and Multi-Threaded. Solver options are accessed from the pull-down menu (refer to Listing 12). The Multi-Threaded solver should be the fastest, with the Advanced solver the second fastest. SAP2000 documentation describes possible numerical instability for "sensitive" structures and analysis methods which seem to be directed at inelastic forms of analysis. If deemed necessary, a spot check could be performed using the Regular solver.



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Error Reading Plot Data: Modal time-history results are typically read using amplitude-frequency plots. This error message appears when there are a lot of output time steps. A large number of steps can result from high speed machines in large structures. One workaround is to limit the quantity of output steps (in the Load Case dialog box); another is to set a limited time range (in the Plot Function Trace dialog box) before requesting a plot.

Converting a RISA-3D model to SAP2000

RISA 3D is very commonly used for the static analysis of steel structures, including modules. Though RISA-3D can perform a modal separation and a response spectrum analysis, the program does not have a dynamic analysis capability suitable for dynamic equipment. SAP2000 is capable of performing a static analysis as well as several forms of analysis suitable for dynamic equipment.

For a static analysis, RISA-3D has a clear advantage over SAP2000 based on ease of use and widespread familiarity.

For a vibratory analysis, SAP2000 needs to be used because RISA-3D has no capability to perform this type of analysis. Typically a modal time-history or steady-state analysis is performed.

Note: Attempts to transfer RISA-3D model data into a SAP2000 file have so far been unsuccessful. Failure to transition data files implies that a SAP2000 model must be input independently of the RISA-3D model.

Selected Specific Listings

There are several ways to implement each of the following tasks. The following listings describe the use of pulldown menus.

Listing 1: Create a Joint Spring:

One or more joints must exist in the model.

- a) Clear the model of selections using the pulldown menu <Select><Clear Selection>. If this menu option is dimmed, then model selections are already cleared.
- b) Select a joint by clicking on the desired joint. The joint should then display with a X shaped selection marker.
- c) Assign spring properties using the pulldown menu <Assign><Joint><Springs>. The resulting dialog box should appear as shown in Figure 1.

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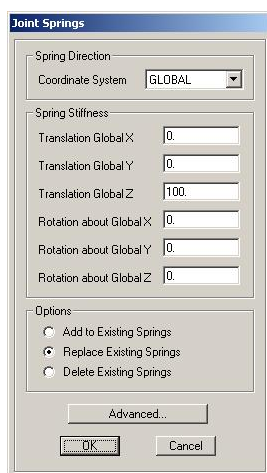


Figure 1: Joint Springs

d) Accept input values by clicking the "OK" button.

Notes: The coordinate system may be switched between local (1, 2, 3 axes) and global (X, Y, Z axes). The advanced option allows the input of cross stiffness values. Stiffness units of measure are not displayed, the user has to know what units to use. Refer to the following:

<u>Direction</u>	<u>Units</u>	<u>Example</u>
translation stiffness	force/length	kN/m
rotational stiffness	force-length/angle	kN-m/rad
cross stiffness	force/angle	kN/rad

Assigned spring values can then be reviewed by right-clicking on the desired node.

Listing 2: Create a (1 Joint) Support Element:

One or more joints must exist in the model.

- Create a set of support element properties using the pulldown menu <Define><Section Properties...><Link/Support Properties>. Click on the "Add New Property..." button. The resulting dialog box should appear as shown in Figure 2a.

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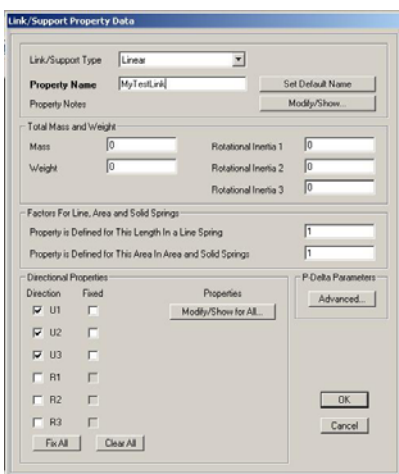


Figure 2a: Link/Support Property Data

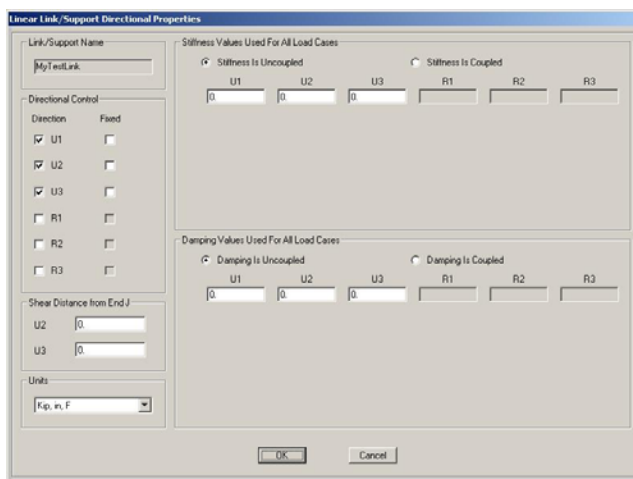


Figure 2b: Linear Link/Support Directional Properties

- b) Enter a name and desired properties into the dialog box. The axes for spring stiffness and damping are set by checking the appropriate axis boxes in the lower portion of the dialog box. Values for the checked axes are entered by first clicking the "Modify/Show for All.." button. The resulting dialog box should appear as shown in Figure 2b.

Stiffness and damping units of measure are not displayed; the user has to know what units to use. Refer to Listing 1 for stiffness units and the following for damping units:

<u>Direction</u>	<u>Units</u>	<u>Example</u>
translation damping	force-time/length	kN-s/m
rotational damping	force-length-time/radians	kN-m-s/rad
cross damping	force-time/radians	kN-s/rad

Support elements are input using local axis orientations. For a default orientation, the translation to global axis orientation is as follows:

<u>Local Axis</u>	<u>Corresponding Global Axis</u>
U1	Z (upward)
U2	X
U3	Y

- c) Accept input values by clicking the "OK" button for each dialog box.
- d) Assign the support element properties to a node using the pulldown menu <Draw><Draw 1 Joint Link>. In the Properties of Object dialog box, leave the offsets at zero. Then click on the desired node or nodes. To close the dialog box, press <Escape>.

Link assignments can be reviewed by using the pulldown menu <View><Set Display Options...> and then checking the Link Properties checkbox.

NON-RIGID SUPPORT FOR VIBRATORY EQUIPMENT

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Listing 3: Create a (2 Joint) Link Element:

Two or more joints must exist in the model.

- a) Create and define link element properties using steps a, b, and c from Listing 2.
- b) Assign the link element properties to a node using the pulldown menu <Draw><Draw 2 Joint Link>. In the Properties of Object dialog box, leave the offset at zero. Then click on the desired 2 nodes and then right click at an open location. Repeat the clicking process until all desired 2 joint links are assigned. To close the dialog box, press the "Escape" button.

Link elements are input using local axis orientations. Local axis orientations are similar to those for frame elements.

Link assignments can be reviewed by using the pulldown menu <View><Set Display Options...> and then checking the Link Properties checkbox.

Listing 4: Define and Assign Frequency Dependent Properties:

Frequency dependent properties are assigned to link or support elements. Thus a link or support element must be created first.

- a) Create and define frequency dependent properties using the pulldown menu <Define><Section Properties><Frequency Dep Link Props...>. Click the "Add New Property..." button. The resulting dialog box should appear as shown in Figure 3. Enter a name and desired properties into the dialog box.

Frequency	Stiffness (Re)	Damping (Im)
Hz	Kip/in	Kip/in
1	0	0
2		
3		

Figure 3: Frequency Dependent Link Property Data

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- b) Enter a name and desired properties into the dialog box. Separate properties are entered for each degree of freedom. The degree of freedom pulldown box is used to change from one DOF to another.
- b) Accept input values by pressing the "OK" button.
- c) Clear the model of selections using pulldown menu <Select><Clear Selection>. If this menu option is dimmed, then model selections are already cleared. Then select previously created link or support properties using the pulldown menu <Select><Select<>Properties><Link Properties>. From the Select Properties dialog box, choose the desired link or support property and press the "OK" button.
- d) Assign frequency dependent properties using the pulldown menu <Assign><Link/Support><Frequency Dep Link Props...>. From the Assign/Define Frequency Dependent Links dialog box, choose the frequency dependent property to be applied. Then accept the choice by pressing the "OK" button.

Frequency dependent property assignments can be reviewed by using the pulldown menu <View><Set Display Options...> and then checking the Link Properties checkbox.

Listing 5: View or Modify Member Stiffness Properties:

One or more frame members must exist in the model.

- a) Clear the model of selections using pulldown menu <Select><Clear Selection>. If this menu option is dimmed, then model selections are already cleared. Then select by clicking on each intended members to be modified. Selected members will be displayed as dashed.
- b) Assign stiffness modifiers using the pulldown menu <Assign><Frame><Property Modifiers...>. The resulting dialog box should appear as shown in Figure 4.

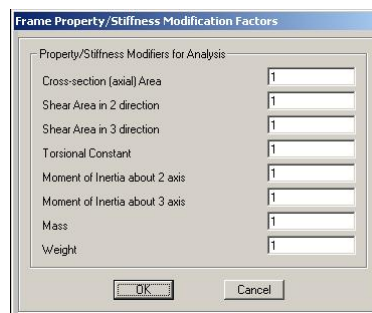


Figure 4: Frame Property/Stiffness Modification Factors

- c) Enter the desired modifiers. These are treated as multipliers, so a value of 1.0 provides no modification.
- d) Accept the input modifiers by pressing the "OK" button.

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Listing 6: View or Modify Load Case Damping:

One or more dynamic load cases must exist in the model. For a forced response analysis, this would be either a time-history case or a steady-state case.

- a) Access the load case using the pulldown menu <Define><LoadCases...>. The resulting dialog box should appear as shown in Figure 5.

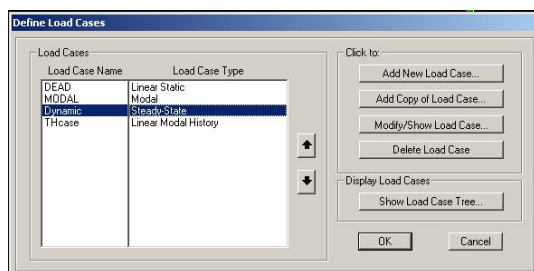


Figure 5: Define Load Cases

- b) From the Define Load Cases dialog box, choose a load case of type "Steady-State" or "Linear Modal History" from the list and then press the "Modify/Show Load Case" button. The resulting dialog box should be similar to that shown in Figure 6.

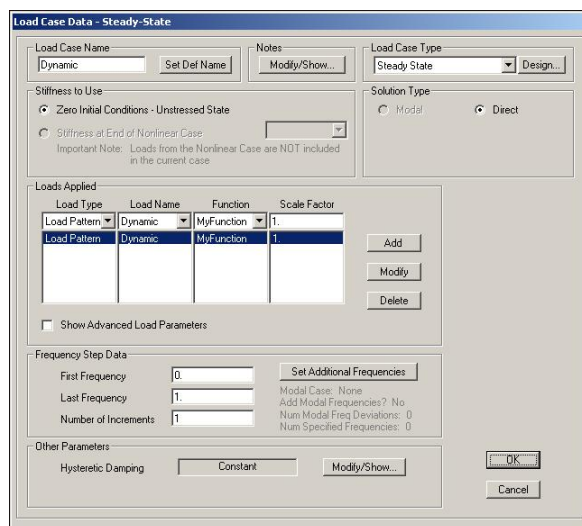


Figure 6: Load Case Data - Steady-State

- c) Access damping input by pressing the "Modify/Show..." button. The details of this screen and input values will depend on the type of dynamic analysis being performed.
- d) Accept damping input by pressing the "OK" button.

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Listing 7: View or Modify Material-Specific Damping

One or more material definitions must exist in the model.

- Access the material list using the pulldown menu <Define><Materials...>. Press the "Show Advanced Properties" button.
- From the Define Materials dialog box, choose the material to be modified and then press the "Modify/Show Material..." button.
- From the Material Property Options dialog box, press the "Modify/Show Material Properties..." button.
- From the Material Property Data dialog box, press the "Material Damping Properties..." button. The resulting dialog box should appear as shown in Figure 7.

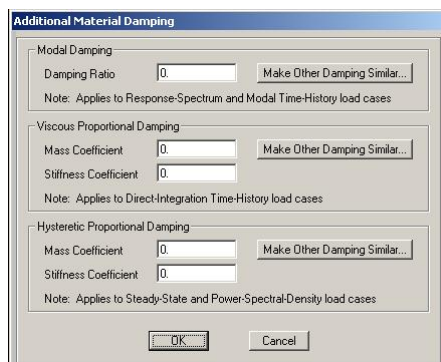


Figure 7: Additional Material Damping

Three types of material damping input are indicated. Dialog box notes indicate which type of damping input applies to which forms of dynamic analysis.

- Accept damping input by pressing the "OK" button.

Listing 8: Directly Specify and Assign Mass at a Joint:

One or more joints must exist in the model.

- Clear the model of selections using pulldown menu <Select><Clear Selection>. If this menu option is dimmed, then model selections are already cleared. Then select by clicking on each node to be modified. Selected nodes will be marked with an X.
- Assign joint mass by using the pulldown menu <Assign><Joint><Masses...>. The resulting dialog box should appear as shown in Figure 8.

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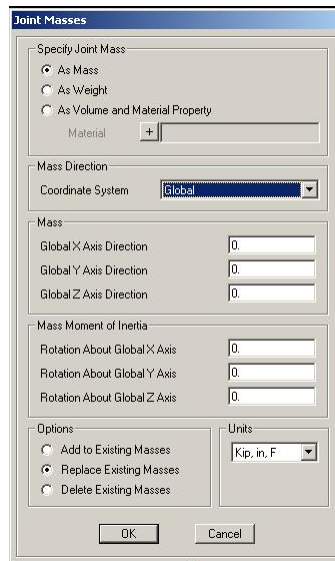


Figure 8: Joint Masses

Notes: The coordinate system may be switched between local (1, 2, 3 axes) and global (X, Y, Z axes). The units of measure depend on mass option chosen at the top of the dialog box.

- c) Accept mass input by pressing the "OK" button.

Assigned mass values can then be reviewed by right-clicking on the desired node.

Listing 9: Specify that Mass be Computed from a Load Pattern,

One or more load patterns must exist in the model.

- a) Define mass source data through the pulldown menu <Define><Mass Source...>. The resulting dialog box should appear as shown in Figure 9.

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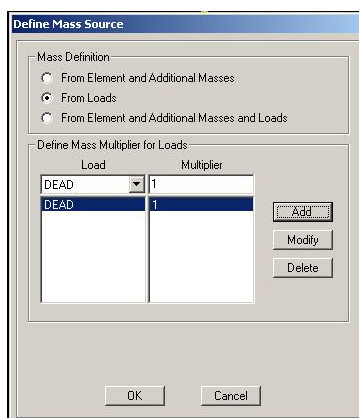


Figure 9: Define Mass Source

- b) Then select either the "From Loads" or "From Element and Additional Masses and Loads" option. Select one or more load patterns from the list. Apply a multiplier for each load pattern as applicable. For example, if 20% snow load is to be included as mass, enter 0.2.
- c) Accept mass input by pressing the "OK" button.

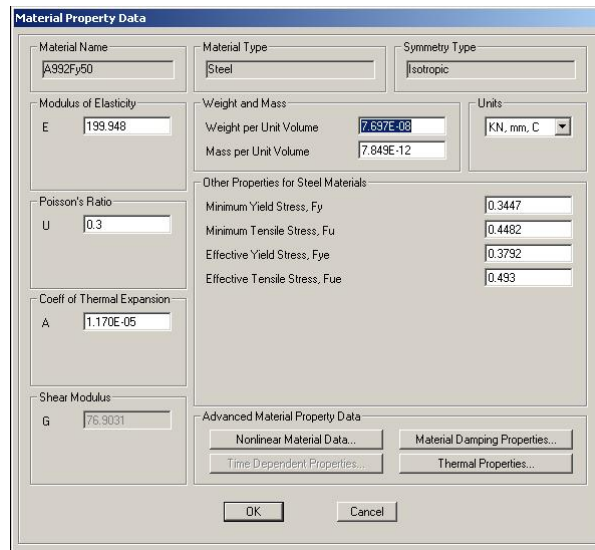
Listing 10: View or Modify Material Mass Density

One or more materials must exist in the model.

- a) Access the material definition using the pulldown menu <Define><Materials...>. The Define Materials dialog box should appear.
- b) Click the "Show Advanced Properties" checkbox.
- c) Choose a material on the left side of the dialog box.
- d) Press the "Modify/Show Material Properties" button. The resulting dialog box should appear as shown in Figure 10.
- e) The value in the "Mass per Unit Volume" may then be changed. The units of measure are mass/volume (i.e. kg/m³).

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The 'Material Property Data' dialog box is shown with the following fields and values:

- Material Name:** A992Fy50
- Material Type:** Steel
- Symmetry Type:** Isotropic
- Modulus of Elasticity (E):** 199,948
- Poisson's Ratio (U):** 0.3
- Coeff of Thermal Expansion (A):** 1.170E-05
- Shear Modulus (G):** 76,9031
- Weight and Mass:**
 - Weight per Unit Volume:** 7.697E-08
 - Mass per Unit Volume:** 7.849E-12
 - Units:** KN, mm, C
- Other Properties for Steel Materials:**
 - Minimum Yield Stress, Fy:** 0.3447
 - Minimum Tensile Stress, Fu:** 0.4482
 - Effective Yield Stress, Fye:** 0.3792
 - Effective Tensile Stress, Fue:** 0.493
- Advanced Material Property Data:**
 - Nonlinear Material Data...
 - Material Damping Properties...
 - Time Dependent Properties...
 - Thermal Properties...

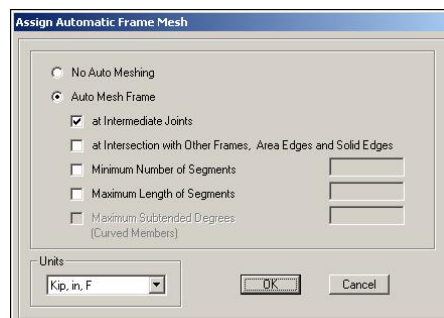
Buttons: OK, Cancel

Figure 10: Material Property Data

Listing 11: Assign Automatic Meshing to a Frame Member:

One or more frame members must first exist

- Clear the model of selections using pulldown menu <Select><Clear Selection>. If this menu option is dimmed, then model selections are already cleared. Then select by clicking on each member to be modified. Selected members will be marked as dashed.
- Assign mesh properties using the pulldown menu <Assign><Frame><Automatic Frame Mesh...>. The resulting dialog box should appear as shown in Figure 11.



The 'Assign Automatic Frame Mesh' dialog box is shown with the following options and values:

- Radio Buttons:**
 - ☐ No Auto Meshing
 - ☒ Auto Mesh Frame
- Checkboxes:**
 - ☒ at Intermediate Joints
 - ☐ at Intersection with Other Frames, Area Edges and Solid Edges
 - ☐ Minimum Number of Segments
 - ☐ Maximum Length of Segments
 - ☐ Maximum Subtended Degrees (Curved Members)
- Units:** Kip, in, F

Buttons: OK, Cancel

Figure 11: Assign Automatic Frame Mesh

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c) To limit the maximum size of an element, check the "Maximum Length of Segments" checkbox and fill in the associated data field with a length.

d) Accept mesh input by pressing the "OK" button.

Assigned mesh properties can then be reviewed by right-clicking on the desired joint.

Listing 12: Access Solver Options

a) Access solver options using the pulldown menu <Analyze - Set Analysis Options...>. Then press the "Solver Options..." button. The resulting dialog box should appear as shown in Figure 12.

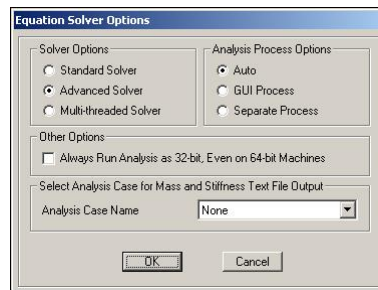


Figure 12: Equation Solver Options

b) From the Equation Solver Options dialog box, make the desired choices.

c) Accept choices by pressing the "OK" button.

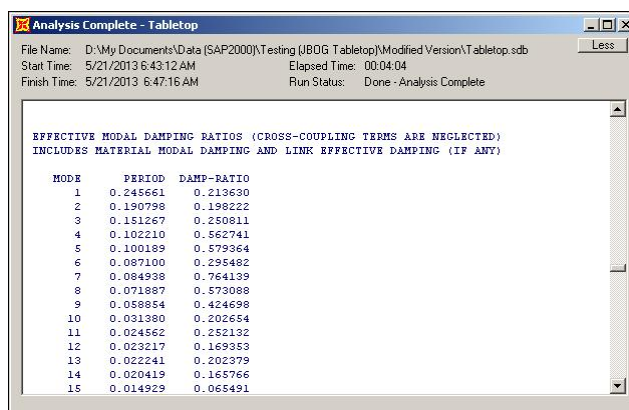
Listing 13: Retrieve Modal Results (Damping)

A modal time-history load case must first be run.

To access modal damping values, use the pulldown menu <Analyze><Show Last Run Details...>. The "Analysis Complete - ##" box will appear as shown in Figure 13. Scroll down until the modal damping ratio values are visible.

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Analysis Complete - Tabletop

File Name: D:\My Documents\Data (SAP2000)\Testing (JB0G Tabletop)\Modified Version\Tabletop.sdb
Start Time: 5/21/2013 6:43:12 AM Elapsed Time: 00:04:04
Finish Time: 5/21/2013 6:47:16 AM Run Status: Done - Analysis Complete

EFFECTIVE MODAL DAMPING RATIOS (CROSS-COUPLING TERMS ARE NEGLECTED)
INCLUDES MATERIAL MODAL DAMPING AND LINK EFFECTIVE DAMPING (IF ANY)

MODE	PERIOD	DAMP-RATIO
1	0.245661	0.213630
2	0.190798	0.198222
3	0.181267	0.250811
4	0.102210	0.562741
5	0.100189	0.579364
6	0.087100	0.295482
7	0.084938	0.764139
8	0.071887	0.573088
9	0.058854	0.424698
10	0.031380	0.202654
11	0.024562	0.252132
12	0.023217	0.169353
13	0.022241	0.202379
14	0.020419	0.165766
15	0.014929	0.065491

Figure 13: Modal Damping Values

Listing 14: Retrieve Modal Results (Animation)

A modal load case must first be run.

The steps to create in internal animation of a mode shape are very similar to that for a steady-state analysis. Refer to Listing 18.

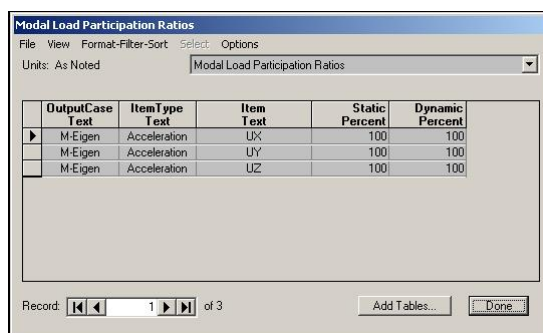
Listing 15: Retrieve Modal Results (Tabular)

A modal load case must first be run.

- To set table options, use the dropdown menu <Display><Tables...>. The resulting dialog box should appear similar to that shown in Figure 21.
- Expand the results list for "Analysis Results" and "Structure Output".
- Click the checkbox next to "Modal Information"
- Press the "OK" button to display the table information. The resulting screen should appear similar to that shown in Figure 14.

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OutputCase	Item Type	Item Text	Static Percent	Dynamic Percent
M-Eigen	Acceleration	UX	100	100
M-Eigen	Acceleration	UY	100	100
M-Eigen	Acceleration	UZ	100	100

Figure 14: Modal Tabular Results

The dropdown box near the upper right corner may be used to switch between the different types of results.

Listing 16: Retrieve Time-History Results (Animation)

A modal time-history analysis must first be run.

Option 1: Within the SAP2000 Window

The steps to create an internal animation for a time-history analysis are very similar to that for a steady-state analysis. Refer to Listing 18.

Because a time-history analysis is not restricted to be periodic, linear, or limited to a single frequency, displaying accurate animation results can be data intensive. Thus an on-screen SAP2000 animation plot is a simplification of the computed time-history.

What SAP2000 actually animates is based on specific joint deflection values selected by the user (either max/min, or values at a specific time). For example, if the structure is initially at rest, setting the time to zero will result in no animation movement. However setting the time to another value where there is a deflection, will result in animation movement. Another indicator is that displayed animations do not include the transient portion of the response.

Option 2: External Video

This option creates an accurate representation of the computed time-history; however the resulting avi file size can be large.

- a) To set video options, use the pulldown menu <File><Create Video...><Create Multi-Step Animation Video...>. The resulting dialog box should appear as shown in Figure 15.

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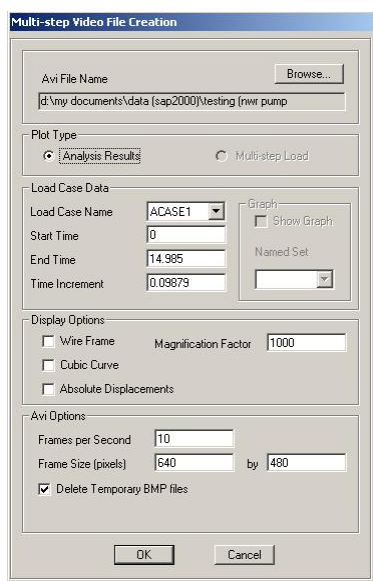


Figure 15: Multi-Step Video File Creation

- b) In the Load Case Name dropdown box, select the appropriate time-history load case. Enter a magnification factor large enough to be visually noticeable.
- e) Press OK to continue.
- f) A second dialog box will appear for data compression options. Edit as necessary and press "OK" to continue.
- f) To run the video, use the Windows Explorer and go to the folder containing the SAP2000 data files. Double click on the avi file to launch the video.

Listing 17: Retrieve Time-History Results (Graphical)

A modal time-history analysis must first be run.

- a) Set plotting options using the pulldown menu <Display><Show Plot Functions...>. The resulting dialog box should appear as shown in Figure 16.

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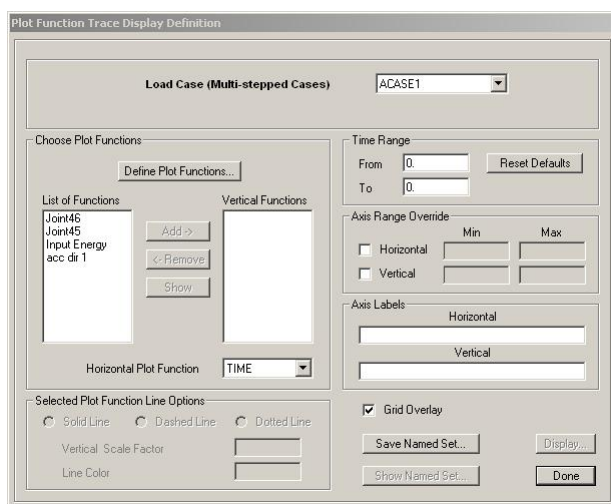


Figure 16: Plot Function Trace Display Definition

- b) The list under "List of Functions" indicates which joints are available for plotting. If the desired point is on the list, select it and press the "Add ->" button.

If the desired joint is not on the list, then press the "Define Plot Functions" button. The resulting dialog box should appear as shown in Figure 17.

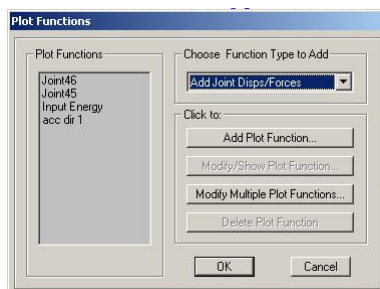


Figure 17: Plot Functions

In the dropdown box, select "Add Joint Disps/Forces", and then press the "Add Plot Function..." button. The resulting dialog box should appear as shown in Figure 18.

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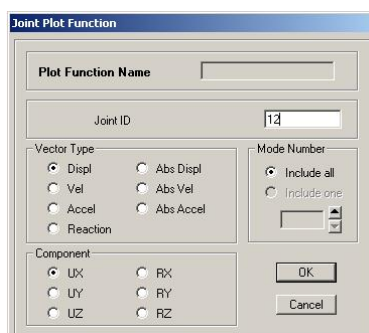


Figure 18: Joint Plot Function

Enter the desired joint number in the edit box; select vector type = "Displ"; select the desired component direction.

To accept the entry, press the "OK" button twice to get back to the Plot Function Trace Display Definition.

- c) Press the "Display..." button. The resulting trace should appear similar to that shown in Figure 19a.

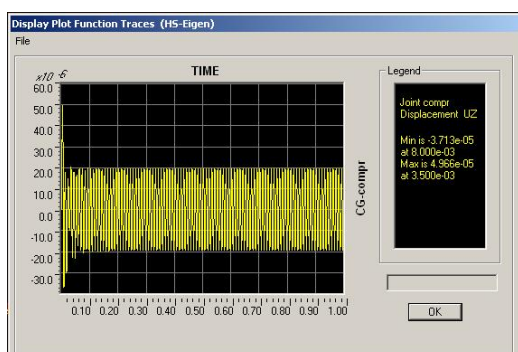


Figure 19a:

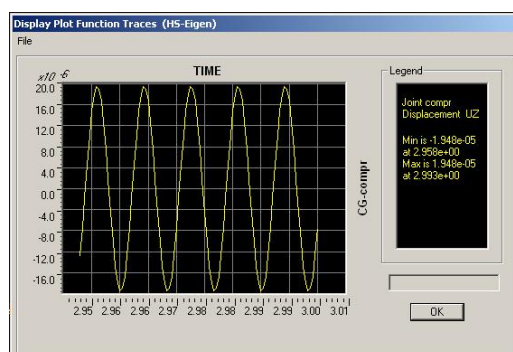


Figure 19b:

Note: There is no need to retrieve peak amplitudes by interpolating the vertical scale. Maximum and minimum values are provided on the right side of the window. If a transient response appears in the graph (Figure 19a), then the start time can be reset to a time value after the transient has disappeared. Then the displayed maximum and minimum values will correspond to the steady-state portion of the response (Figure 19b).

Note: Plot values can be saved to a text file using the dialog box pulldown menu <File><Print Tables to File...>.

Listing 18: Retrieve Steady-State Results (Animation)

One or more steady-state analyses must first be run.

Because a steady-state analysis is restricted to harmonic functions, the data required to represent an accurate animation is relatively simple. Thus an internal SAP2000 steady-state animation accurately represents the computed

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SAP2000 Implementation Notes

time dependent deflections.

- a) To set deformation options, use the pulldown menu <Display><Deformed Shape...>. The resulting dialog box should appear as shown in Figure 20.

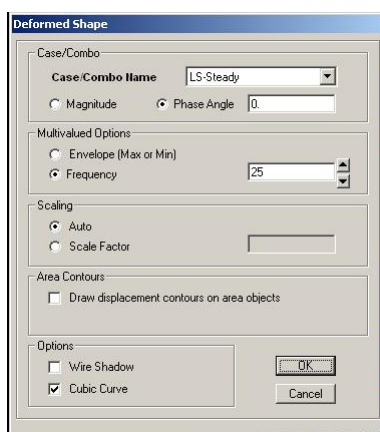


Figure 20: Deformed Shape

- b) In the Case Combo Name dropdown box, select the appropriate steady-state load case. Select "magnitude" to show the peak magnitude, instead of the magnitude at phase angle zero. Select the desired frequency, if a range was part of the steady-state analysis.
- e) Press OK to continue. A structure deflected shape should appear on the screen.
- f) Press the "Start Animation" button at bottom of SAP2000 window.

Listing 19: Retrieve Steady-State Results (Tabular)

One or more steady-state analyses must first be run.

- a) To set table options, use the dropdown menu <Display><Tables...>. The resulting dialog box should appear as similar to that shown in Figure 21.

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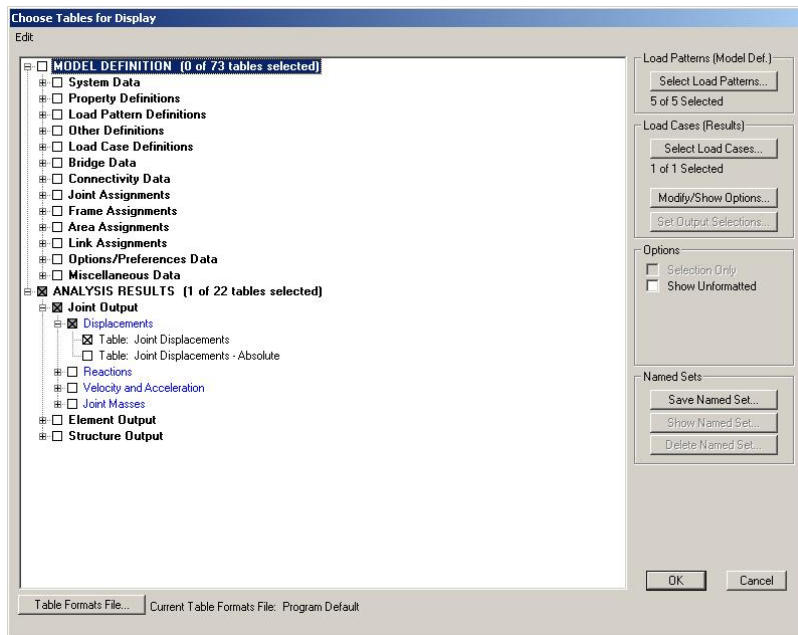


Figure 21: Choose Tables for Display

- Click the box to the left of "joint output" and "displacements" to expand and "Table - Joint Displacements" to select as indicated in Figure 21.
- Press the "Select Load Cases..." button and edit as desired.
- Press the "Modify/Show Options..." button. The resulting dialog box should appear as shown in Figure 22.

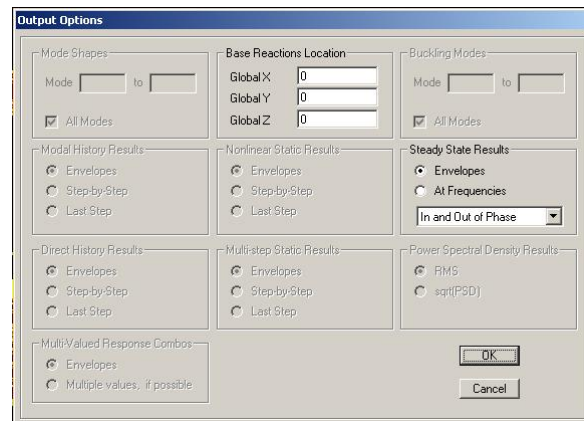


Figure 22: Output Options

- Select either "envelopes" or "at frequencies". Select "in and out of phase", "magnitude", or "all". The option for

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"in and out of phase" will provide the real and imaginary components of the dynamic amplitude. The "magnitude" will provide the modulus.

- f) Press OK to close output options
- g) Press OK to display the table.
- h) Filter for a specific node using the dialog box dropdown menu <Format-Filter-Sort><Filter Table>.
- i) Select "Quick", specify field = joint, operator = "=", Value = desired node
- j) Press OK to display the filtered table.

Note: Data may be extracted by using the dialog box pulldown menu <File><Export Current Table><To Excel> to store displayed table data to a Microsoft Excel file.